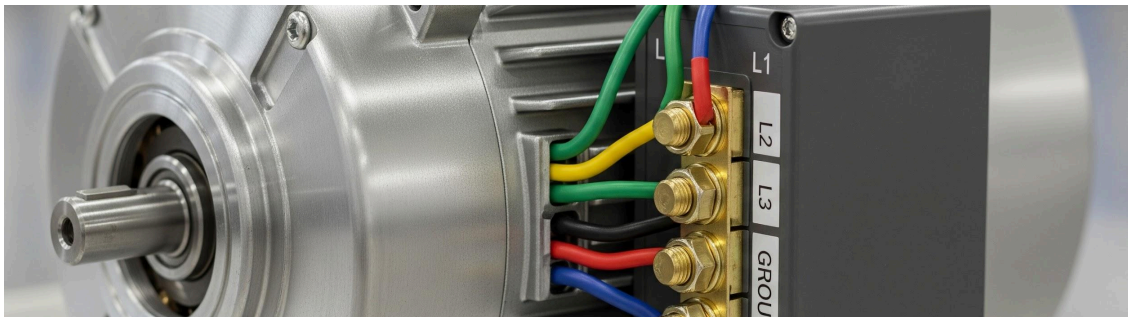

Electrical Machine Data Sheet

Machine Name/Type : Synchronous Motor

Name plate Details

The rating section details the machine's operational capacity and design limits.

Parameter	Value	Unit
Power/Output	5	HP
Voltage	415	V
EXTN. Volts	220	V
Amps	8.2	A
Amps	1.55	A
Frequency	50	Hz
Phase	3	Phase
Speed	1500	RPM

Precautions

I. Ensure the supply voltage is **415 V AC, 3-phase** before operation.

II. **Do not apply mechanical load before synchronizing** the motor with supply.

III. Field excitation must be applied **only after the motor reaches near synchronous speed**.

IV. Never switch OFF field excitation while the motor is running under load.

V. Verify correct phase sequence before starting.

VI. Ensure proper synchronization to avoid heavy current and mechanical shock.

VII. Maintain proper earthing of the motor body.

VIII. Do not exceed rated excitation current.

IX. Monitor temperature, vibration, and sound during operation.

X. Switch OFF immediately if hunting, abnormal noise, or overheating is observed.

Working Principle of Synchronous Motor

A synchronous motor operates on the principle of **magnetic locking between rotating and stationary magnetic fields**.

When a three-phase AC supply is given to the stator, a **rotating magnetic field** is produced.
The rotor is excited with **DC supply**, producing a constant magnetic field.

When rotor speed reaches synchronous speed, the rotor magnetic field locks with the stator field, and the motor runs at **constant speed equal to synchronous speed**, independent of load.

Working Characteristics of Synchronous Motor

I. Runs at constant speed, regardless of load variations.

II. Speed is equal to synchronous speed only.

III. Not self-starting; requires external starting methods.

IV. Power factor can be controlled by varying field excitation.

V. Highly efficient for constant-speed applications.

VI. Can operate at lagging, unity, or leading power factor.

Applications

1. Power factor improvement in industries
2. Constant-speed drives
3. Large compressors
4. Pumps and blowers
5. Conveyors
6. Rolling mills
7. Crushers
8. Cement plants
9. Textile industries
10. Industrial power systems